(8) 1. Poarta CMOS: SAU-NU

a) funcționare, caracteristica statică de transfer, marg. de zgomot

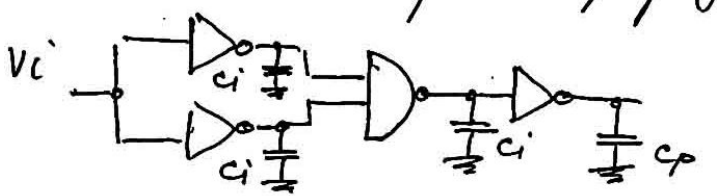
b) Să se calc. timpul de propagare pt o poartă fără tampon.

$C_p = 50\,pF$, $R_{TN} = R_{Tp} = 1k\Omega$; $V_{DD} = 5^V$, $V_T = 2{,}5^V$

c) Să se calc. timpul de propagare pt o poartă cu tampon (SAU-NU)

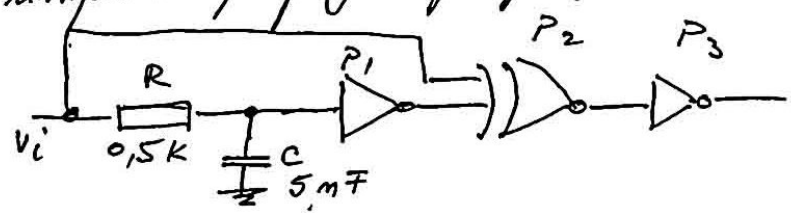
$C_i = 5\,pF$, $C_p = 50\,pF$

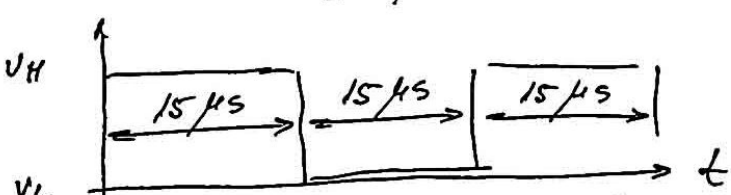
$R_{TN} = R_{Tp} = 1k\Omega$



2. a) Se dă circ. de mai jos. Se cere să se ridice diagramele de timp de funcționare în punctele A, B, C, D.

b) Să se determine valoarea timpilor rezultați la ieșire. Vor fi neglijați timpii de propagare pe porți





$V_{CC} = 5^V$; $V_H = 3{,}5^V$, $V_L = 0{,}2^V$; $V_T = 1{,}5^V$

SUBIECT D

D1. Poarta CMOS : SAU-NU

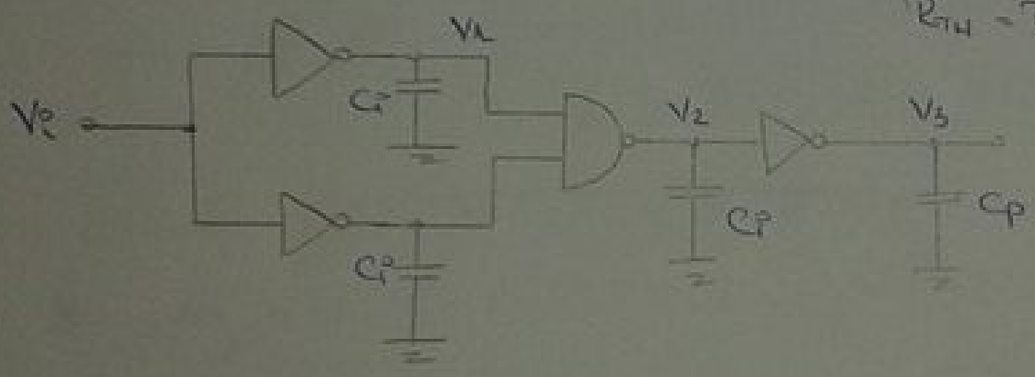
a) funcționare, caracteristică statică de transfer, marginea de zgomot.

b) să se calculeze timpul de propagare pt. o poartă fără tampon : $V_{DD} = 5V$, $V_T = 2{,}5V$, $C_p = 50\,pF$, $R_{TN} = R_{TP} = 1k\Omega$

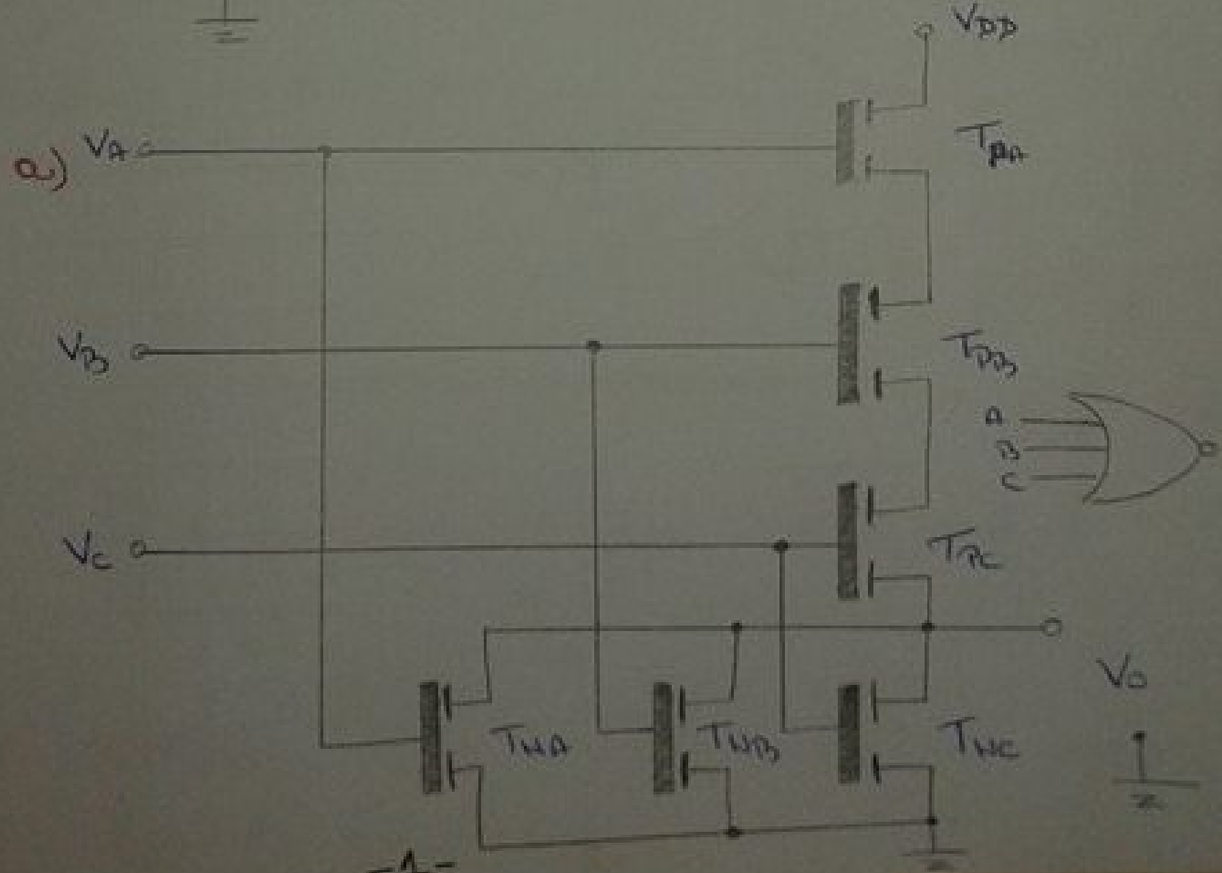
c) să se calculeze timpul de propagare pt. o poartă cu tampon (SAU-NU)

$C_i = 5pF$; $C_p = 50pF$

$R_{TN} = R_{TP} = 1k\Omega$



a)



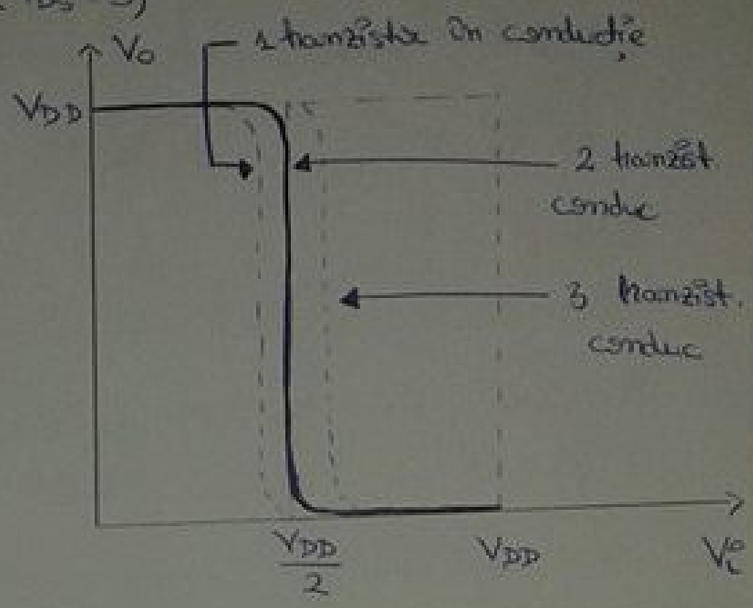
- $V_A = V_B = V_C = V_H$

T_{HA}, T_{HB}, T_{HC} - conduc

T_{PA}, T_{PB}, T_{PC} - blocați

$\Rightarrow V_O = 0V\ (V_L)$

($i_{DS} = 0$)



- $V_A = V_B = V_C = V_L$

T_{PA}, T_{PB}, T_{PC} - conduc

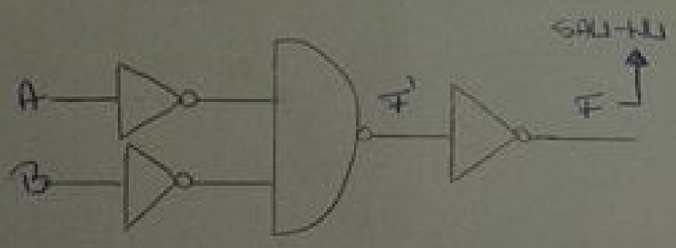
T_{HA}, T_{HB}, T_{HC} - blocați

$\Rightarrow V_O = V_{DD} - V_H$

$F = \overline{A + B + C}$

$V_O = V_{THx} + \overset{=0}{V_{GND}} \Rightarrow$ cu cât mai multe T_H, scade rezistența deci și $V_{THx} \Rightarrow$ scade V_O

- un tranzistor CMOS la ieșire $\Rightarrow$ un singur tranzistor NMOS conduce (ori PMOS ori NMOS de la ieșire indiferent de intrări)



$\overline{F'} = \overline{\overline{A} \cdot \overline{B}}$

$F = \overline{\overline{A} \cdot \overline{B}} = \overline{\overline{\overline{A} + \overline{B}}} = \overline{A + B}$

- Tabel de funcționare

A	B	C	F
L	L	L	H
L	L	H	L
L	H	L	L
L	H	H	L
H	L	L	L
H	L	H	L
H	H	L	L
H	H	H	L

• Margine de zgomot

$$M_z^+ = V_{OH\,min} - V_{IH\,min} = \frac{1}{3} V_{DD} \quad \left(\text{ideal finde la } \frac{V_{DD}}{2}\right)$$

$$M_z^+ = V_{DD}\left(\frac{1}{1,5 + \frac{m_i}{m_c}} - 0,01\right)$$

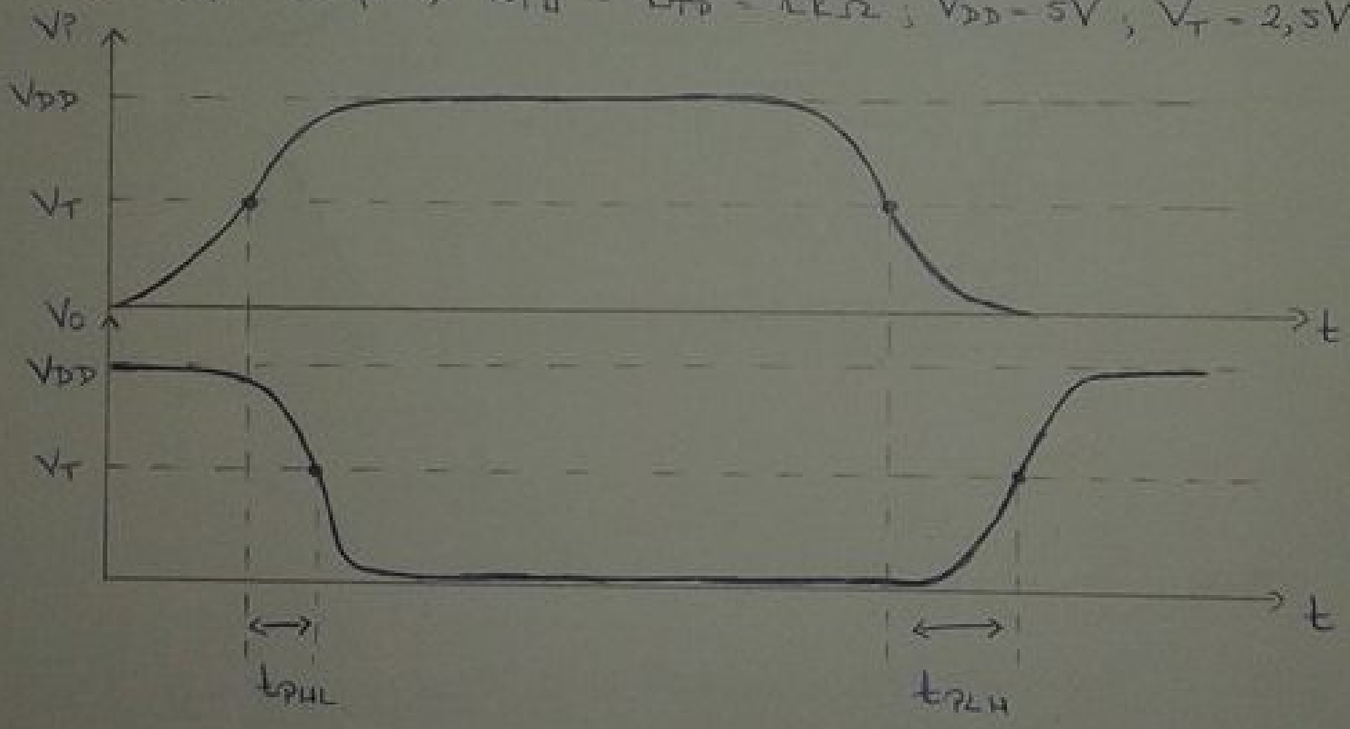
$$M_z^- = V_{IL\,max} - V_{OL\,max} = \frac{1}{3} V_{DD} \quad \left(\text{ideal finde la } \frac{V_{DD}}{2}\right)$$

$$M_z^- = V_{DD}\left(0,99 - \frac{1}{1,5 + \frac{m_i}{m_c}}\right)$$

m_i - nr. de inverzori utilizate pe poartă

m_c - nr de inverzori total pe poartă

b) $C_p = 50pF$; $R_{TH} = R_{TD} = 1k\Omega$; $V_{DD} = 5V$; $V_T = 2,5V$

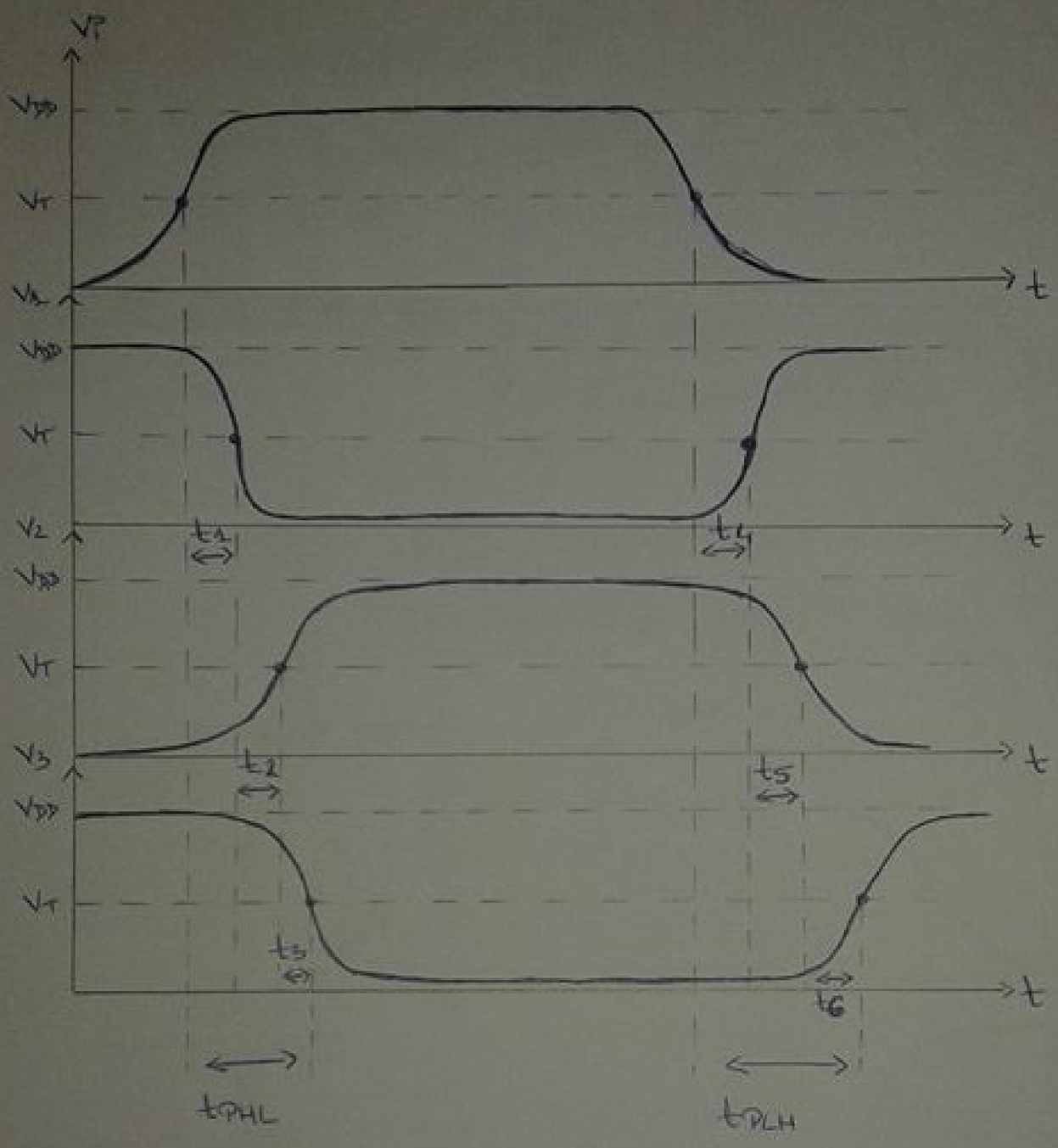


$$t_{PHL} = \frac{R_{TH}}{2} C_p \ln 2 = \frac{1 \cdot 10^3}{2} \cdot 50 \cdot 10^{-12}\, 0,7 = 17,5\ ns$$

$$t_{PLH} = 2 R_{TD} \cdot C_p \ln 2 = 2 \cdot 1 \cdot 10^3 \cdot 50 \cdot 10^{-12} \cdot 0,7 = 70\ ns$$

$$t_{PD} = \frac{t_{PHL} + t_{PLH}}{2} = 43,75\ ns$$

c) $C_p = 50\,pF$; $C_i = 5\,pF$; $R_{TN} = R_{TP} = 1\,k\Omega$



$t_{PHL} = t_1 + t_2 + t_3$

$t_1 = R_{TN} \cdot C_i \cdot \ln 2 = 1 \cdot 10^3 \cdot 5 \cdot 10^{-12} \cdot 0{,}7 = 3{,}5\,ns$

$t_2 = \frac{R_{TP}}{2} \cdot C_i \cdot \ln 2 = \frac{1 \cdot 10^3}{2} \cdot 5 \cdot 10^{-12} \cdot 0{,}7 = 1{,}75\,ns$

$t_3 = R_{TN} \cdot C_p \cdot \ln 2 = 1 \cdot 10^3 \cdot 50 \cdot 10^{-12} \cdot 0{,}7 = 35\,ns$

$\Rightarrow t_{PHL} = 40{,}25\,ns$

$t_4 = R_{TP} \cdot C_i^o \cdot \ln 2 = 1 \cdot 10^3 \cdot 5 \cdot 10^{-12} \cdot 0,7 = 3,5 ms$

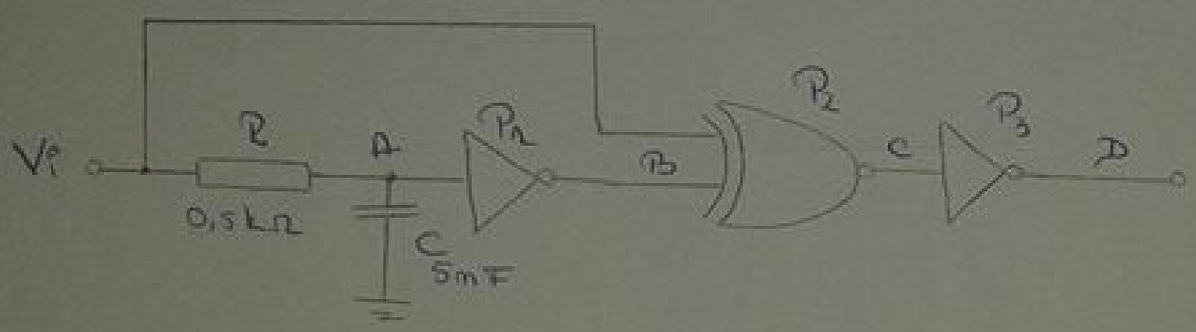
$t_5 = 2 R_{TH} \; C_i^o \cdot \ln 2 = 2 \cdot 1 \cdot 10^3 \cdot 5 \cdot 10^{-12} \cdot 0,7 = 7 ms$

$t_6 = R_{TP} \cdot C_p \cdot \ln 2 = 1 \cdot 10^3 \cdot 50 \cdot 10^{-12} = 35 ms$

$t_{PLH} = t_4 + t_5 + t_6 = 45,5 ms$

$\Rightarrow t_{PD} = \dfrac{t_{PHL} + t_{PLH}}{2} = 43,8 ms$

D2.



$V_{cc} = 5V$; $V_H = 3,5V$; $V_L = 0,2V$; $V_T = 1,5V$

$\tau = R \cdot C = 0,5 \cdot 10^3 \cdot 5 \cdot 10^{-9} = 0,5 \cdot 5 \cdot 10^{-6} =$

$= 2,5 \mu s < 15 \mu s$

$T = 30 \mu s$

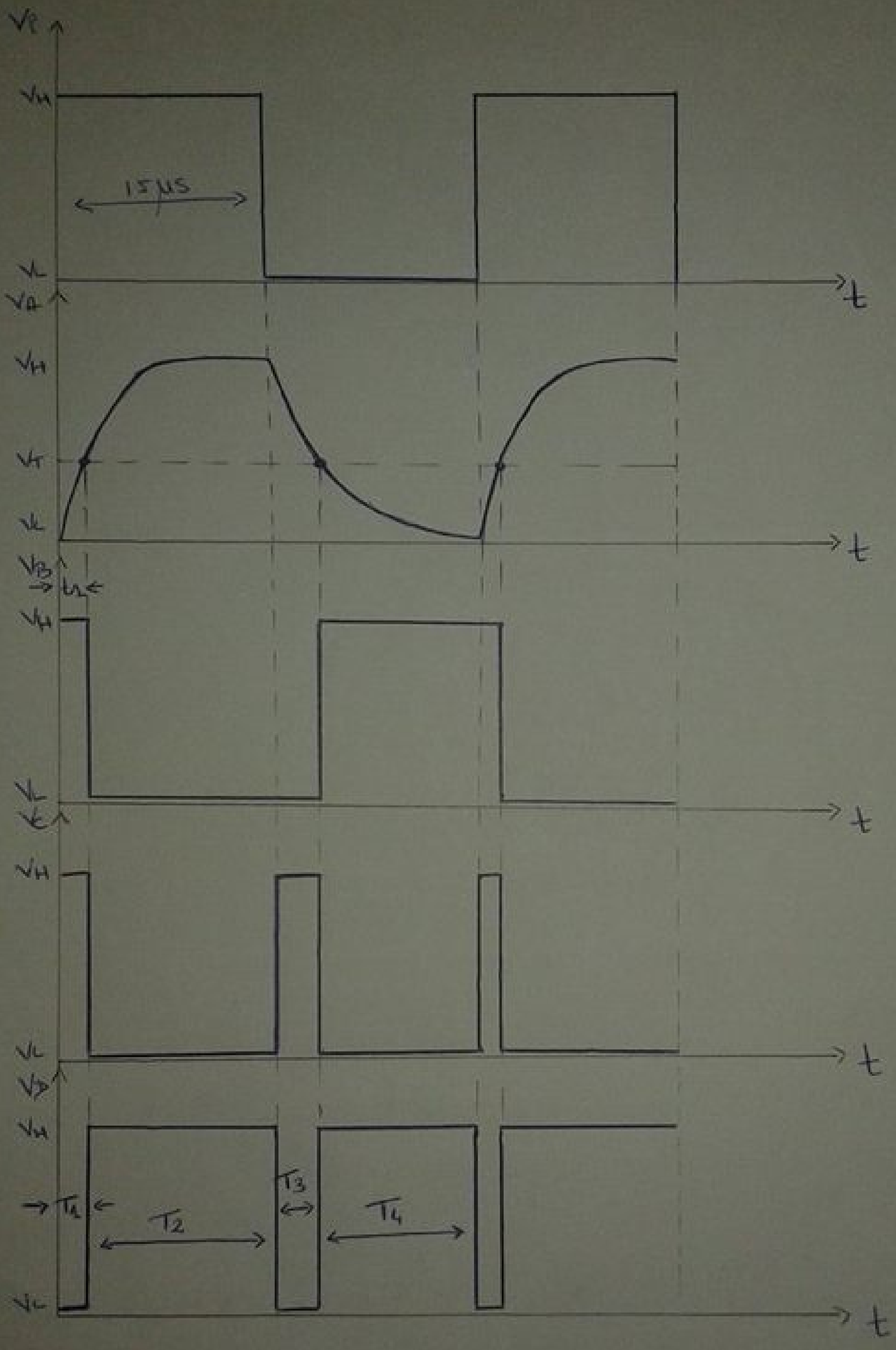
$t_1 = R \cdot C \cdot \ln \dfrac{V_H - V_L}{V_H - V_T} = 2,5 \cdot 10^{-6} \ln \dfrac{3,5 - 0,2}{3,5 - 1,5} =$

$= 2,5 \cdot 10^{-6} \cdot \ln \dfrac{2,3}{2} = 1,25 \mu s$

$\Rightarrow T_1 = 1,25 \mu s \Rightarrow T_2 = 15 \mu s - 1,25 \mu s = 13,75 \mu s$

$T_3 = 2,32 \mu s$ $\left(\text{dim } t_3 = R \cdot C \cdot \ln \dfrac{V_L - V_H}{V_L - V_T} = \dots = 2,32 \mu s \right)$

$\Rightarrow T_4 = 15 \mu s - 2,32 \mu s = 12,68 \mu s$


VP
VH
15 µS
VL
t
VA
VH
VT
VL
t
VB
t2
VH
VL
t
VC
VH
VL
t
VD
VH
T1
T2
T3
T4
VL
t